

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses correct formula to obtain an expression for the magnitude of the resultant force or the acceleration	AO3.4	M1	Force towards centre of circle = $\frac{990v^2}{48}$
	Forms a correct equation involving friction	AO1.1b	A1	Friction = $\frac{990v^2}{48}$
	Deduces the value of F to be used and substitutes to find the maximum safe speed in m s^{-1}	AO2.2a	M1	$v = \sqrt{\frac{48(7300)}{990}}$
	Obtains the correct maximum safe speed converting to miles per hour AWR40	AO1.1b	A1	$v = 18.8(13..) \text{ m s}^{-1}$ $v = 42 \text{ mph}$
(b)	Infers that on a wet day friction would be reduced	AO2.2b	E1	Wet conditions reduce friction
	Infers that the 10000 N is incorrect and concludes that Gary's revised assumption is wrong	AO2.2b	E1	10000 N > 9200 N Gary's assumption is wrong
				Total 6 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	250 N
Total 1 mark			

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Resolves vertically to form an equation to find the tension.	AO3.1b	M1	$0.4 \times 9.8 = T \cos 30^\circ$
	Finds correct tension.	AO1.1b	A1	$T = 4.5 \text{ N}$
(b)	Finds radius.	AO1.1b	B1	$r = 0.6 \sin 30^\circ$ $= 0.3$
	Forms an equation to find	AO3.1b	M1	$T = \sin 30^\circ = 0.4 \times r\omega^2$

	angular speed with 'their' radius and 'their' tension.			
	Obtains correct angular speed to 2 sf. FT 'their' values provided both M1 marks have been awarded	AO1.1b	A1F	$\omega = \sqrt{\frac{4.5 \sin 30^\circ}{0.4 \times 0.3}}$ $= 4.3 \text{ rad s}^{-1}$
(c)	States two appropriate assumptions.	AO3.5b	B1	Light and inextensible.
Total 6 marks				

Q4.

- (a) Resolve vertically:

$$T \cos \theta = mg$$

M1 for $T \cos \theta$ or $T \sin \theta$ and mg

M1

$$34 \cos \theta = 2 \times 9.8$$

A1

$$\cos \theta = \frac{19.6}{34}$$

$$\theta = 54.8^\circ$$

A1

3

- (b) Resolve horizontally for particle:

$$\frac{mv^2}{r} = T \sin \theta$$

M1 for $T \cos \theta$ or $T \sin \theta$

M1

$$v^2 = \frac{34 \sin 54.8^\circ \times 0.8}{2}$$

A1 ft
from
(a)

$$v^2 = 11.113$$

Speed is 3.33 m s^{-1}

Accept 3.34

A1

3

(c) Time taken is $2\pi r / v$

Or find ω and use $\frac{2\pi}{\omega}$

M1

$$= 1.51 \text{ sec}$$

A1ft

2

[8]

Q5.

In limiting equilibrium, using $F = \mu R$

Frictional force is $0.2 \times mg$

M1A1

Resolve horizontally

$$\frac{m \times 15^2}{r} = 0.2 \times mg$$

M1

$$r = \frac{15^2}{0.2 \times g}$$

$$= 114.79$$

$$= 115$$

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A1

[4]

Q6.

$$R = mg$$

M1

$$F = 0.85 mg$$

A1

$$\frac{mv^2}{r} = 0.85 mg$$

$$\text{condone } \frac{mv^2}{r} = 0.85R \text{ (for M1A1)}$$

M1A1

$$v^2 = 34 \times 0.85 \times g$$

$$= 283.22$$

dependent on both M1s

m1

$$v = 16.8 \text{ m s}^{-1}$$

A1

[6]

Q7.

- (a) For particle B,
tension in string = $2.1g$ N

B1

Resolve horizontally for particle A:

$$m\omega^2 r = T$$

$$\text{Or } m_1 \omega^2 r = m_2 g \text{ or } \frac{m_1 v^2}{r} = m_2 g$$
(condone lack of 1 and 2)

M1

$$1.4\omega^2 \times 0.3 = 2.1g$$

$$\omega^2 = 49$$

A1

Angular velocity is 7 rad / sec

A1

4

- (b) Using $v = r \omega$:
speed = 0.3×7

M1

$$= 2.1 \text{ m s}^{-1}$$

Part (b) marks can be awarded in (a)

A1

2

- (c) Time taken is $2 / \omega$

$$\text{Or } \frac{2\pi r}{2.1}$$

M1

$$= \frac{2\pi}{7} = 0.898 \text{ sec}$$

Accept $\frac{2\pi}{7}$
(0.895 M1A0)

A1

2

[8]

Q8.

- (a) *Resolve vertically* $R = mg$

If the particle is on the point of sliding, $F = \mu R$

Ignore all inequalities

M1

$$\therefore F = 0.3R = 0.3mg$$

A1

Resolving radially: $F = m\omega^2 r$

M1

$$0.3mg = m\omega^2 \times 0.8$$

$$\omega^2 = \frac{0.3 \times g}{0.8}$$

$$\omega = 1.92$$

A1

4

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- (b) (i) 45 revolutions per minute = $\frac{90\pi}{60}$

M1

$$= \frac{3\pi}{2} \text{ or } 4.71 \text{ radians per second}$$

A1

2

- (ii) *Resolving radially:* $F = m\omega^2 r$

$$m \mu g = m \left(\frac{3\pi}{2} \right)^2 \times 0.15$$

M1A1 either side correct

A1

A1 second side correct

$$\mu = \frac{\left(\frac{3\pi}{2}\right)^2 0.15}{g}$$

$$\mu = 0.340$$

CAO (accept 0.339)

A1

4

[10]

Q9.

(a) Resolving vertically

$$T \cos 30 + 20 \cos 50 = 4g$$

M1: Three terms, which must include $4g$, $T \cos \theta$ or $T \sin \theta$ and $20 \cos \theta$ or $20 \sin \theta$, where $\theta = 30, 40, 50$ or 60 .

A1: Correct terms

A1: Correct equation

M1A1
A1

$$T \cos 30 = 26.344$$

$$T = 30.4 \text{ N}$$

A1: Correct final answer.

Accept 30.4 or AWRT 30.42.

Accept 30.4 or 30.5 or AWRT 30.45 from $g = 9.81$.

A1

4

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(b) Horizontally: $\frac{mv^2}{r} = 20 \cos 40 + T \cos 60$

M1: Three terms, which must include $\frac{mv^2}{r}$ or $\frac{4 \times 5^2}{r}$, $T \cos \theta$ or $T \sin \theta$ and $20 \cos \theta$ or $20 \sin \theta$, where $\theta = 30, 40, 50$ or 60 .

A1F: Correct equation. May include T, m and v .

M1
A1F

$$\frac{4 \times 5^2}{r} = 30.53$$

dM1: Substitution of values for T, m and v .

Equation of form $\frac{4 \times 5^2}{r} = \text{number}$

dM1

$$r = 3.27537$$

$$= 3.28$$

A1: Correct answer. Accept 3.27 or 3.28 or AWR 3.28.
Accept 3.27 or AWR 3.27 from $g = 9.81$.

$$\frac{mv^2}{r}$$

Note: Do not accept $\frac{mv^2}{r} = 30.4$ **or similar.**

A1

4

[8]

Q10.

As particle moves, $T = \frac{mv^2}{r}$

M1

If radius is r , extension is $r - 1.2$

B1

Using $T = \frac{\lambda x}{l}$:

$$T = \frac{192(r - 1.2)}{1.2}$$

M1

$$= 160(r - 1.2)$$

A1

$$T = \frac{mv^2}{r} \Rightarrow 160(r - 1.2) = \frac{8 \times 3^2}{r}$$

M1

$$160r^2 - 192r = 72$$

$$(\text{or } 192r^2 - 230.4r = 86.4)$$

A1

$$20r^2 - 24r - 9 = 0$$

$$(10r + 3)(2r - 3) = 0$$

M1

$$r = 1.5 \text{ or } -0.3$$

Radius is 1.5

A1

or
using unknown as extension:

If extension is x , radius is $1.2 + x$ B1

$$T = \frac{\lambda x}{l}$$

Using

$$T = \frac{192x}{1.2} \quad \text{M1}$$

$$= 160x \quad \text{A1}$$

$$T = \frac{mv^2}{r} \Rightarrow 160x = \frac{8 \times 3^2}{1.2 + x} \quad \text{M1M1}$$

$$192x + 160x^2 = 72 \quad \text{A1}$$

$$20x^2 + 24x - 9 = 0$$

$$(10x - 3)(2x + 3) = 0 \quad \text{M1}$$

$$x = 0.3 \text{ or } -1.5$$

$$\text{Radius is } 1.5 \quad \text{A1}$$

[8]

Q11.

(a) $a = \frac{14^2}{50} = 3.92$

finding acceleration

M1

correct acceleration

A1

$$F = 1200 \times 3.92 \quad \text{AG}$$

use of $F = ma$

dM1

$$= 4704 \text{ N}$$

correct force from correct working

A1

4

(b) $R = 1200 \times 9.8 = 11760$

normal reaction

B1

$$4704 \leq \mu \times 11760$$

applying $F \leq \mu R$

M1

$$\mu \geq \frac{4704}{11760}$$

AG

correct result from correct working

A1

3

$$\mu \leq 0.4$$

[7]

Q12.

(a) (i) Acceleration = $\frac{v^2}{r} = \frac{(7.5)^2}{15}$
Attempt at $\frac{v^2}{r}$

M1

$$= 3.75 \text{ ms}^{-2}$$

A1

2

(ii) $2940 = 400 \times \frac{v^2}{15}$
M1 use, A1 subs correct

M1A1

$$V = 10.5 \text{ ms}^{-1}$$

A1

3

(b) Motorcycle and rider modelled as a particle

B1

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 Size of rider/cycle compared with radius / 15 m

B1

2

(c) Acceleration of force $\left(\frac{v^2}{r}\right)$ must decrease
Force decrease → radius increase B1 sc

M1

so r must increase

For 2 marks, algebraic reference or convincing explanation

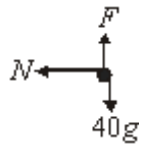
A1

2

[9]

Q13.

(a)



F towards centre scores B0

B1

1

(b) (i) $F = 40g$ or 392 N

B1

1

(ii) $F \leq \mu N$

$$392 \leq \mu 784$$

use of $\leq$ or $=$

M1

$$\mu \geq \frac{392}{784} = 0.5$$

A1

2

(iii) $N = m r \omega^2$

$m r \omega^2$ seen or used

M1

$$784 = 40 (3) \omega^2$$

values substituted

A1

$$\omega^2 = 6.53$$

$$\omega \approx 2.56$$

AG

A1

3

(c) Martin modelled as a particle
any suitable assumption

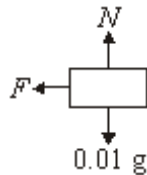
B1

1

[8]

Q14.

(a)



B1

1

(b) $F = mr\omega^2$

Formula quoted and attempt at use

M1

$$= 0.01(0.4)\omega^2$$

$$= 0.004\omega^2$$

A1

2

(c) $F = \mu N$

$$= 0.8 (0.01g)$$

Use of limiting friction

M1

$$= 0.008g$$

or 0.0784 seen

A1

Limiting, so

$$0.004\omega^2 = 0.008g$$

Equates their expressions

M1

$$\omega^2 = 2g$$

$$\omega = 4.43$$

AG

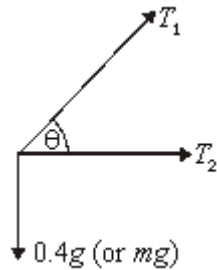
A1

4

[7]

Q15.

(a)



Three forces evident;

T_1, T_2 clear – on diagram or in calculations

B1

1

(b) Vertically $T_1 \sin \theta = 0.4g$

Resolve vertically – component evident

$T \sin \theta = mg$ seen

M1 A1

$$T_1 \left(\frac{4}{5} \right) = 0.4g$$

M1

$$T_1 = 0.5g = 4.9\text{N}$$

AG obtained

A1

3

(c) $T_2 = 0.1k$

B1

1

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(d) $a = r\omega^2 = 0.3 \times 5^2$

Calculation of a seen

B1

$$\text{Force} = T_2 + T_1 \cos \theta$$

(Both terms of horizontal resultant force attempted)

M1

$$= 0.1k + 4.9 \times \frac{3}{5}$$

Previous expression / result substituted

A1

$$F = ma$$

Their ' a ' and ' F ' (Dependent on first M1)

m1

$$k = 0.6$$

AG

A1

5

(e)
$$\text{EPE} = \frac{kx^2}{2} = \frac{0.6(0.1)^2}{2}$$

$$\frac{kx^2}{2} \text{ seen and attempt to use}$$

M1

$$= 0.003 \text{ J}$$

A1

2

[12]

Q16.

(a) 2π rads take 27.32 days

2π rads take $27.32 \times 24 \times 60 \times 60$ sec

days to seconds

B1

$$\text{angular speed} = \frac{2\pi}{(27.32 \times 24 \times 60 \times 60)}$$

radians per second

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$$\approx 2.66 \times 10^{-6} \text{ rad s}^{-1}$$

must see 4sf

A1

3

(b) Using $v = r\omega$:

$$= (3.844 \times 10^8) \times (2.66 \times 10^{-6})$$

use of formula

M1

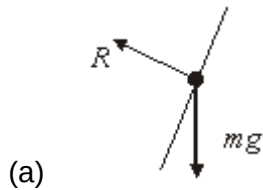
$$\approx 1020 \text{ ms}^{-1} \text{ (3sf)}$$

AWRT

A1

2

Q17.



Correct diagram (to include arrows and labels)

B1

1

(b) $R \cos 60^\circ = mg$

Resolving vertically

M1

$R = 2mg$

ag Correct R from correct working

A1

2

(c) $R \cos 30^\circ = \frac{mv^2}{r}$

Resolving horizontally

M1

Correct equation

A1

$r = \frac{v^2}{g\sqrt{3}}$

Solving for r

m1

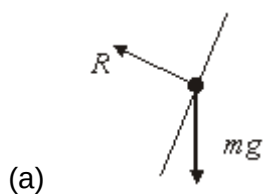
Correct r

A1

4

[7]

Q18.



Correct diagram (to include arrows and labels)

B1

1

- (b) $R \cos 60^\circ = mg$
Resolving vertically

M1

$$R = 2mg$$

ag Correct R from correct working

A1

2

- (c) $R \cos 30^\circ = \frac{mv^2}{r}$
Resolving horizontally

M1

Correct equation

A1

$$r = \frac{v^2}{g\sqrt{3}}$$

Solving for r

m1

Correct r

A1

4

- (d) Decrease to $\frac{1}{4}$ of previous value
Decrease

B1

$$\frac{1}{4}$$

B1

2

[9]

Q19.

- (a) (i) $R \cos 60^\circ = 3 \times 9.8$
Resolving vertically
Correct equation

M1 A1

$$R = 58.8 \text{ N}$$

Correct R

A1

3

(ii) $58.8 \cos 30^\circ = 3 \times \frac{v^2}{0.5}$
Resolving vertically
Correct equation

M1 A1

$$v = \sqrt{\frac{58.8 \cos 30^\circ}{6}} = 2.91 \text{ ms}^{-1}$$

Solving for v
Correct v

M1 A1

4

(b) (i) No change
No change

B1

1

(ii) Increased because v^2 is proportional to the radius
Increases
Reason

B1 B1

2

[10]

EXAM PAPERS PRACTICE

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Examiner reports

Q1.

Students demonstrated a good understanding of circular motion and the associated forces. The mean mark for part (a) was 66% and the mean mark for part (b) was 69%.

In part (a) students demonstrated a clear understanding of the formula $F = \frac{mv^2}{r}$. Marks were subsequently lost either by using the incorrect value for friction — either 9200 N or an average of the two stated values — or incorrectly converting speed from metres per second. It is well worth students thinking about how reasonable their final answer is to pinpoint any errors; for example speeds of 40000 mph or 0.5 mph are clearly not valid.

In part (b) the best answers clearly stated that ‘wet conditions reduce friction’ and that ‘Gary’s assumption was wrong as the upper limit had increased’. For both marks to be awarded, students needed to explain why the friction values should have been reduced and highlight the fact that the upper limit had increased in the proposed model.

Q4.

In part (a) many students resolved vertically to find the angle but in the required equation $T \cos \theta = mg$ a number of students used $\sin \theta$ rather than $\cos \theta$ whilst others used $mg = \frac{T}{\cos \theta}$. Similar errors were seen in part (b). In part (c) many students found ω and then stopped. The value of ω was not credited since it was just as simple to find the time from knowing only v as it was from knowing ω .

Q5.

This question was answered well by most students. A few students were concerned that the mass of the particle was not given in the question even though the m ’s cancelled in the equation which the students were required to use.

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Q6.

This question was answered succinctly by the better candidates. Others were concerned that the mass was not given, and some tried to include terms in $mg \cos \theta$. Some

found $\frac{mv^2}{r} = 0.85R$ but were unable to find R .

Q7.

Many candidates resolved horizontally for particle A and obtained $m\omega^2 r = mg$ without appreciating that the m in $m\omega^2 r$ was 1.4, being the mass of particle A, whereas the m in mg was 2.1, being the mass of particle B. Other candidates found the speed, v , to be 2.1ms⁻¹ but never found ω , or even included any equation involving ω .

Q8.

There were many good solutions to this question. A common problem seen in part (a) was candidates stating that $F = \mu R$ but being unable to notice that $R = mg$. Virtually all candidates

could convert 45 revolutions per minute into an angular speed of $\frac{3\pi}{2}$ radians per second. Part (c) was usually completed well.

Q9.

Many candidates answered this question well. Part (a) was usually correct but a few candidates only considered the component of one of the tensions to equate to mg . Most candidates answered part (b) well.

Q10.

Some candidates assumed that energy needed to be considered. Those who did equate

tension to be $m \frac{v^2}{r}$ sometimes found $m \frac{v^2}{r} = \frac{\lambda x}{1.2}$ but then often failed to notice that $r = 1.2 + x$, which reduced their equation to one with only one unknown.

Q11.

Part (a) was done very well by the vast majority of the candidates. The printed answer probably helped some of them. In part (b), many candidates were able to obtain the value of 0.4, but far fewer were able to deal correctly with the inequality. When attempting questions like this candidates should start with $F < \mu R$ and preserve the inequality through out their working. Typically no attempt was made to start with an inequality, which some candidates did then try to introduce at the end of their solution.

Q12.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q13.

There was an excellent response to this question – far better than similar questions in the past - with many candidates scoring 7 or 8 marks. Quite often the force diagram proved problematic with friction incorrectly pointing to the centre of the circle or an additional force existing opposite to the normal reaction. The use of the law of friction was good. Not all candidates were able to give a modelling assumption that was not already stated in the stem. Most common and correct answers here were “treat Martin as a particle” or “air resistance ignored”.

Q14.

This question had a mixed response. In part (a), many candidates did not draw a correct force diagram, usually showing a fourth force or ignoring friction altogether. The structure of the question was there to help candidates, but some ignored the structure when answering parts (b) and (c). Full marks were duly awarded for this approach.

Q15.

This proved to be a demanding question although a significant proportion of candidates solved it well. The force diagram in part (a) was rarely correct with equal tensions shown or, even worse, the tension in OP missing and sometimes a normal reaction appearing at P . Those candidates who had not clearly marked different tensions were able to recover the mark if their working out indicated the use of two different tensions.

In parts (b) and (d) many candidates struggled to set up the equations required. The answer in part (b) was arrived at through incorrect methods to match the answer given. In part (d) mrw^2 was often seen, but a number of candidates thought that this was directed along QP or outwards from P .

Parts (c) and (e) were more successful and it was pleasing to see the correct formulae in use.

Q16.

This was a good starter. Many candidates clearly knew the relevant formulae well. The accuracy policy meant that many candidates dropped a mark for failing to show a fourth figure in their final calculation of the angular speed.

Q17.

In part (a), many candidates included an additional force, usually directed towards or away from the centre of the circle. There were many good solutions to part (b). The candidates seemed to have been helped by the printed answer. There were however some incorrect solutions which usually began with a statement such as $R = mg \cos 60^\circ$ and concluded with the printed answer. The candidates found part (c) more difficult, but there were still a good number of correct solutions. A few candidates left $\cos 30^\circ$ as part of their answer and did not substitute a value for this.

Q18.

In part (a), many candidates included an additional force, usually directed towards or away from the centre of the circle. There were many good solutions to part (b). candidates seemed to have been helped by the printed answer. There were however some incorrect solutions which usually began with a statement such as $R = mg \cos 60^\circ$ and concluded with the printed answer. The candidates found part (c) more difficult, but there were still a good number of correct solutions. A few candidates left $\cos 30^\circ$ as part of their answer and did not substitute a value for this.

Q19.

In part (a), almost all of the candidates produced the value of 58.8 N, but quite a few of these did not give a satisfactory argument to support their answer. Some candidates were clearly dividing the weight of the particle by the radius of the circle. In part (b), the common error was to simply use 58.8 without resolving to find the horizontal component. The answers to part (b) were very varied. Some candidates did produce reasonable answers without attempting part (a).